

# LPS Tier 2 — binary-path correctness gates: E2.14 (separation robustness) and E2.12 (selection-metric consistency)

## Contents

|                                                                    |          |
|--------------------------------------------------------------------|----------|
| <b>Purpose</b>                                                     | <b>1</b> |
| <b>E2.14 — the constructed separation fixtures</b>                 | <b>2</b> |
| <b>E2.14 — deviance trajectories: plain Newton vs step-halving</b> | <b>3</b> |
| <b>E2.12 — the constructed G6 fixture</b>                          | <b>4</b> |
| <b>E2.12(a) — selection scores the deployed clipped metric</b>     | <b>5</b> |
| <b>E2.12(b) — the log-loss clip: one point owned the score</b>     | <b>6</b> |
| <b>E2.12 study — selection stability across truncation levels</b>  | <b>7</b> |
| <b>Results summary and discussion</b>                              | <b>8</b> |
| <b>What we learned</b>                                             | <b>9</b> |
| <b>Appendix — reproducibility</b>                                  | <b>9</b> |

**Report build:** 2026-06-11 10:20:31 EDT

**Result generation:** 2026-06-11 10:20:19 EDT (UTC 2026-06-11T14:20:19Z)

**Source commit:** cb89e12e7c19 (branch codex/geosmooth-t2-binary-hygiene)

**Report source (repo-relative):** dev/methods/lps/reports/tier2/binary\_hygiene/2026-06-11/scripts/lps\_tier2\_binary\_hygiene\_report.Rmd

**Input manifest:** inputs/tier2\_report\_inputs\_manifest.txt

**Bundle cross-check:** ok vs audit\_artifacts/tier2\_20260611T084642Z: e2\_12a max |diff| = 4.441e-16; e2\_14 traces max |diff| = 1.421e-14

All numbers shown are read from precomputed, committed CSV artifacts — the report-specific inputs in `inputs/` beside this report (regenerated deterministically by the adjacent `scripts/export_tier2_report_inputs.R` from the gated fixture constructors) and the gate/study evidence at the package-level `reports/` directory; this report did not rerun the experiments. The acceptance evidence proper lives in the execution bundles `audit_artifacts/tier2_20260611T075822Z` (E2.14, commit 75c1788) and `audit_artifacts/tier2_20260611T084642Z` (E2.12 + E2.14 rerun, commit 3eaf9cf), whose regenerated aggregates the input manifest cross-checks.

## Purpose

`fit.lps()` fits local weighted polynomial regression in local charts and selects its configuration (support size, polynomial degree, kernel) by cross-validation. Its binary path has two modes: `bernoulli` (local least squares on 0/1 labels; deployed probabilities are the predictions clipped to  $[0, 1]$ ) and `binomial` (local weighted logistic fits solved by iteratively reweighted least squares, IRLS; selection by cross-validated log loss).

This report documents the results of the two Tier-2 correctness gates that landed on branch `codex/geosmooth-t2-binary-hygiene`:

1. **E2.14 — separation robustness.** Is the local logistic IRLS monotone and bounded on (near-)separable supports, and does *exact* separation end in a documented, telemetered fallback rather than an unbounded loop or NaN? The fix under test is deviance **step-halving** inside the IRLS loop.
2. **E2.12 — selection-metric consistency and log-loss clipping.**
  - (a) Does bernoulli-mode selection score the **deployed** (clipped) metric rather than raw, unclipped predictions — and can the difference change which candidate is selected? (b) Is the log-loss probability truncation pinned at  $10^{-6}$ , and what instability did the previous  $10^{-15}$  truncation permit? A companion **study** (reported, not gated) asks whether the selected candidate is stable across truncation levels  $\{10^{-6}, 10^{-3}\}$ .

Both gates are deterministic **testthat** batteries on constructed fixtures (no replication), so all results below are single realized values; no uncertainty intervals apply. The committed gates, both passing on the committed tree alongside the full Tier-0 battery inside the bundles named above, are:

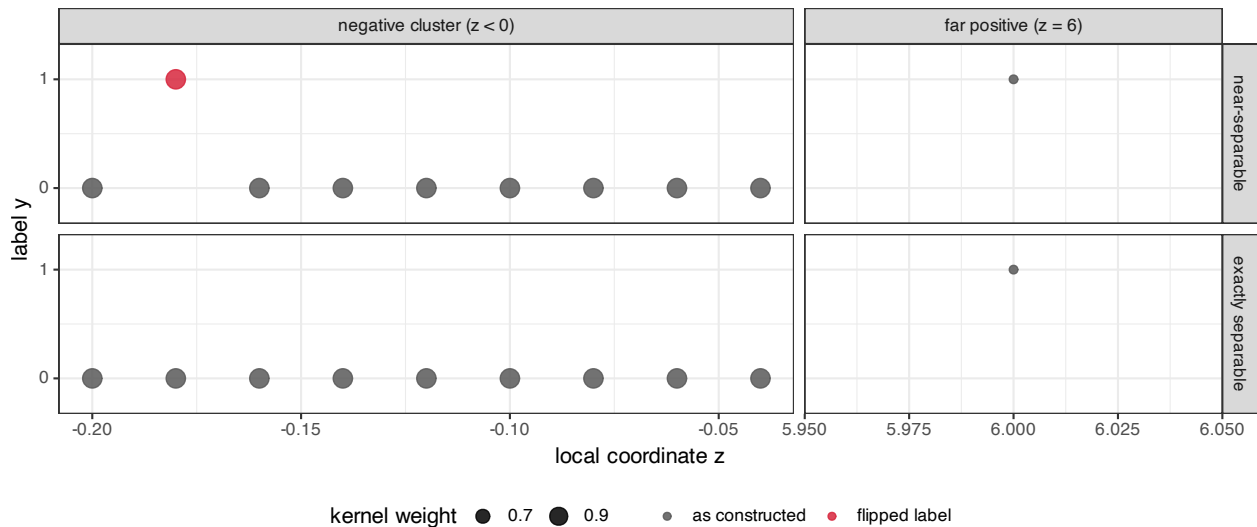
- `tests/testthat/test-lps-binary-separation.R` (E2.14, 43 assertions);
- `tests/testthat/test-lps-binary-metric-consistency.R` (E2.12, 38 assertions).

## E2.14 — the constructed separation fixtures

A local logistic solve receives a small neighborhood: 1-D local coordinates  $z_i$  centered at the evaluation point, labels  $y_i \in \{0, 1\}$ , and gaussian kernel weights  $w_i$  that decay with  $|z_i|$ . The gate uses one support shape in two label configurations (the frozen plan’s DGP:  $y = \mathbf{1}\{z > 0\}$ , optionally one flipped label, weights from the gaussian kernel):

- **Near-separable arm:** ten equispaced negatives  $z \in \{-0.20, -0.18, \dots, -0.02\}$  plus one far positive at  $z = 6$ , with the label of the *second* point ( $z = -0.18$ ) flipped to 1. Two classes, not linearly separable: the flipped positive sits inside the negative cluster.
- **Exactly separable arm:** the same support without the flip — a threshold at any  $z \in (-0.02, 6)$  separates the classes perfectly, so the unpenalized logistic maximum-likelihood estimate does not exist (coefficients diverge).

The flip *position* is load-bearing: flipping an edge-of-cluster point produces the plain-Newton overshoot shown in Figure 2, while a mid-cluster flip leaves plain Newton monotone and would make the fix undetectable.



**Figure 1.** The E2.14 fixture in its two label configurations (rows: near-separable = one flipped label; exactly separable = no flip). The x-axis is split (columns, independent scales) because the far positive at  $z = 6$

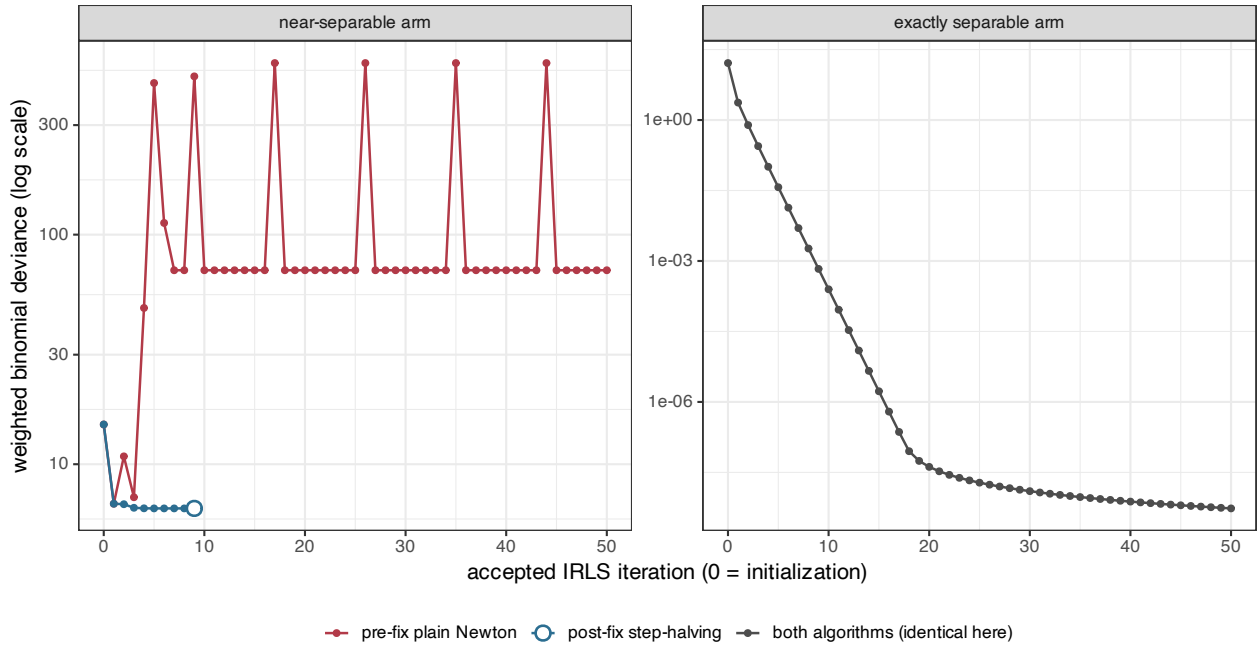
would otherwise compress the negative cluster into a single blot. Point size encodes the gaussian kernel weight: cluster points carry weight  $\approx 1.0$ , the far positive  $\approx 0.61$ . The red point in the top row is the single flipped label at  $z = -0.18$  — a high-weight, wrong-side observation. Removing the flip (bottom row) makes the support exactly separable.

## E2.14 — deviance trajectories: plain Newton vs step-halving

The solver minimizes the weighted binomial deviance

$$D(\beta) = -2 \sum_i w_i \left( y_i \log \mu_i(\beta) + (1 - y_i) \log (1 - \mu_i(\beta)) \right), \quad \mu_i(\beta) = \text{expit}(\tilde{\eta}_i),$$

where  $\tilde{\eta}_i$  is the linear predictor  $\eta_i = x_i^\top \beta$  clamped to  $[-35, 35]$  (the same clamp the IRLS update uses, which keeps  $D$  finite for every finite iterate). Plain IRLS accepts every Newton update unconditionally. The E2.14 fix accepts a candidate  $\beta_{\text{new}}$  only if  $D(\beta_{\text{new}}) \leq D(\beta_{\text{cur}}) + 10^{-8}$ , halving the step toward  $\beta_{\text{cur}}$  up to 30 times otherwise; an exhausted halving budget ends the solve as non-convergent. The gate asserts on the whole recorded trajectory, not only its endpoint, so oscillation cannot hide.



**Figure 2.** Per-iteration deviance of the local logistic solve on the two fixture arms (log scale). *Left:* on the near-separable arm, pre-fix plain Newton (red; replicated update equations of the pre-change solver, committed trajectory in `reports/e2_14_prefix_newton_overshoot.csv`) overshoots at iteration 2 — the deviance jumps from 6.9 to  $\approx 496$ , a  $+489$  increase — then oscillates between  $\approx 6.5$  and  $\approx 9$  and never converges within the 50-iteration cap. Post-fix step-halving (blue) needs a single halving at that same step, descends monotonically (largest per-step increase  $1.8 \times 10^{-15}$ , within the  $10^{-8}$  slack), and converges at iteration 9 (open circle) with finite coefficients and a center prediction  $\hat{p} = 0.125 \in (0, 1)$ . *Right:* on the exactly separable arm both algorithms coincide (no halving triggers): the deviance decreases monotonically toward zero as the coefficients run off to infinity, and the solve stops at the 50-iteration cap as designed.

The realized endpoint quantities behind Figure 2:

Table 1: Post-fix solver summary per fixture arm. Columns: solver status string; whether the coefficient-step criterion (1e-7) was met within the 50-iteration cap; accepted iterations; total step halvings; the largest per-step deviance increase along the accepted trajectory (the gate requires it below 1e-8); the fitted center probability (‘–’ = no converged fit is returned; the fallback layer supplies the prediction, Table 2).

| arm               | status        | converged | iterations | step halvings | max dev. increase | p-hat (center) |
|-------------------|---------------|-----------|------------|---------------|-------------------|----------------|
| near.separable    | ok            | TRUE      | 9          | 1             | 1.78e-15          | 0.1247         |
| exactly.separable | not_converged | FALSE     | 50         | 0             | -1.58e-10         | –              |

On non-convergence the fitting layer falls back deterministically and records it: under `unstable.action = "mean"` the prediction is the local weighted event rate  $\bar{y}_w = \sum_i w_i y_i / \sum_i w_i$ ; under `unstable.action = "na"` it is NA (the sanctioned guarded output). The exactly separable arm exercises both paths:

Table 2: Fallback behavior and telemetry on the exactly separable arm (one solve attempt each). ‘returned value’ is the fitted probability the caller receives: the weighted event rate 0.063151 under “mean”, NA under “na”. The last four columns are the recorded telemetry counters (logistic.diagnostics): every fallback is counted, split into event-rate substitutions and NA returns. No NaN and no unbounded loop occurs.

| unstable.action | returned value | attempted | converged | fallback | event-rate fallback | NA failure |
|-----------------|----------------|-----------|-----------|----------|---------------------|------------|
| mean            | 0.063151       | 1         | 0         | 1        | 1                   | 0          |
| na              | NA             | 1         | 0         | 1        | 0                   | 1          |

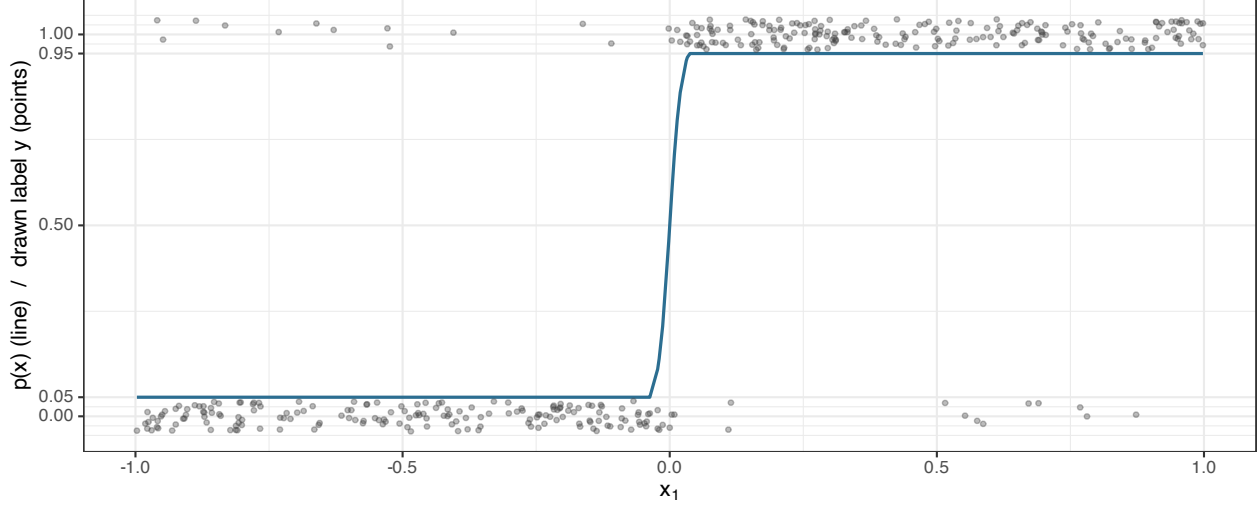
E2.14 also pins that nothing changed where nothing needed to change: when every full Newton step already satisfies the deviance slack, the halving check accepts the identical iterate, so well-behaved solves are numerically identical to the pre-change solver (asserted in the gate against an independent plain-IRLS replication, and corroborated by the Tier-0 E0.6 battery statistics being bit-identical to the unmodified base commit).

## E2.12 — the constructed G6 fixture

Both E2.12 sub-items use the plan’s binary-surface DGP **G6** with one named override: a *sharp* log-odds cliff in place of the default smooth surface,

$$\eta(x) = 6 \tanh(15x_1), \quad p(x) = \min(\max(\text{expit}(\eta(x)), 0.05), 0.95), \quad y \sim \text{Bernoulli}(p),$$

on  $x \in [-1, 1]^2$  with  $n = 400$  (prevalence target 0.5 by symmetry; the  $[0.05, 0.95]$  probability clamp is part of the G6 construction). The cliff at  $x_1 = 0$  is what forces flexible local fits to overshoot  $[0, 1]$  on the raw scale — the behavior sub-item (a) needs. Sub-item (b) uses a second draw (seed 7001) of the same surface with one label deliberately flipped, fit in `binomial` mode.



**Figure 3.** The E2.12(a) fixture (seed 2121): the true success probability  $p(x)$  (blue line; it depends on  $x_1$  only) and the  $n = 400$  drawn labels (grey points, vertically jittered). The surface is a two-plateau cliff —  $p = 0.05$  for  $x_1 \lesssim -0.15$ ,  $p = 0.95$  for  $x_1 \gtrsim 0.15$  — so local polynomial fits near the transition extrapolate outside  $[0, 1]$ . The E2.12(b) fixture is a second draw (seed 7001) of this same surface; its deliberately flipped observation sits deep in the left plateau at  $x_1 = -0.702$ , where every 60-point neighborhood is all-zero.

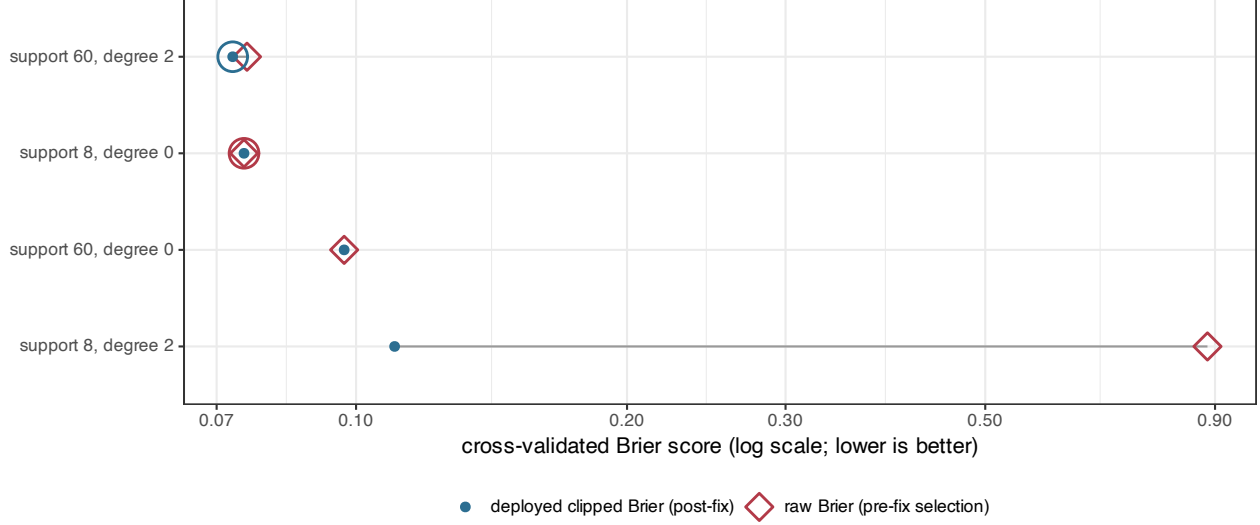
## E2.12(a) — selection scores the deployed clipped metric

For a candidate configuration  $j$  with out-of-fold (cross-validation) predictions  $\hat{p}_{ij}$ , the two scores at issue are

$$\text{raw Brier}_j = \frac{1}{n} \sum_i (y_i - \hat{p}_{ij})^2, \quad \text{deployed Brier}_j = \frac{1}{n} \sum_i (y_i - \text{clip}_{[0,1]}(\hat{p}_{ij}))^2,$$

where  $\text{clip}_{[0,1]}$  is exactly the clamp applied to the probabilities the fitted model actually deploys (`fitted.values`, `predict()`). Pre-fix, bernoulli selection minimized the raw score while deployment clipped — a train/deploy metric mismatch. Post-fix, the selection column `cv.brier.observed` is the deployed score (with `Inf` for any candidate having a non-finite prediction, preserving the old `NA`-handling semantics).

Fit-status accounting for this fixture: all  $4 \times 400 = 1600$  out-of-fold predictions are finite (no `NA` candidates, nothing excluded); 263 of them (16.4%) fall outside  $[0, 1]$ , concentrated in the two degree-2 candidates — the clip is doing real work.



**Figure 4.** Per-candidate selection scores on the E2.12(a) fixture under the two metrics (one row per candidate; open red diamonds = raw Brier, filled blue dots = deployed clipped Brier; grey segments connect a candidate’s two scores; log x-axis; large circles mark each metric’s minimizer). Where a candidate’s predictions never leave  $[0, 1]$  the two scores coincide exactly and the blue dot sits inside the red diamond. Clipping therefore moves nothing for the in-range candidates but rescues the degree-2 candidates whose raw predictions overshoot — most dramatically (support 8, degree 2), whose score falls from 0.883 to 0.110. The two metrics disagree on the winner: the raw score (the pre-fix rule) picks (support 8, degree 0) while the deployed score picks (support 60, degree 2), with cross-candidate margins of  $5.8 \times 10^{-4}$  (raw) and  $2.1 \times 10^{-3}$  (deployed) — both more than  $100\times$  the contract’s  $10^{-6}$  comparison tolerance, so the flip is real rather than a tie. The post-fix fit deploys the clipped winner.

Table 3: The four candidates on the E2.12(a) fixture. ‘raw Brier’ is the mean squared error of the unclipped out-of-fold predictions (the pre-fix selection score); ‘deployed Brier’ is the same on  $[0, 1]$ -clipped predictions and is the post-fix selection column `cv.brier.observed`; the two count columns give each candidate’s out-of-fold predictions outside  $[0, 1]$  (of 400); ‘selected’ marks the candidate the post-fix fit returns. The gate additionally verifies the selection column equals the deployed metric recomputed from the stored CV predictions (max residual  $1.4\text{e-}17$ , contract tolerance  $1\text{e-}6$ ).

| support | degree | raw Brier | deployed Brier | n pred < 0 | n pred > 1 | selected |
|---------|--------|-----------|----------------|------------|------------|----------|
| 8       | 2      | 0.882721  | 0.110365       | 60         | 68         | FALSE    |
| 60      | 0      | 0.097000  | 0.097000       | 0          | 0          | FALSE    |
| 8       | 0      | 0.075078  | 0.075078       | 0          | 28         | FALSE    |
| 60      | 2      | 0.075653  | 0.072937       | 43         | 64         | TRUE     |

## E2.12(b) — the log-loss clip: one point owned the score

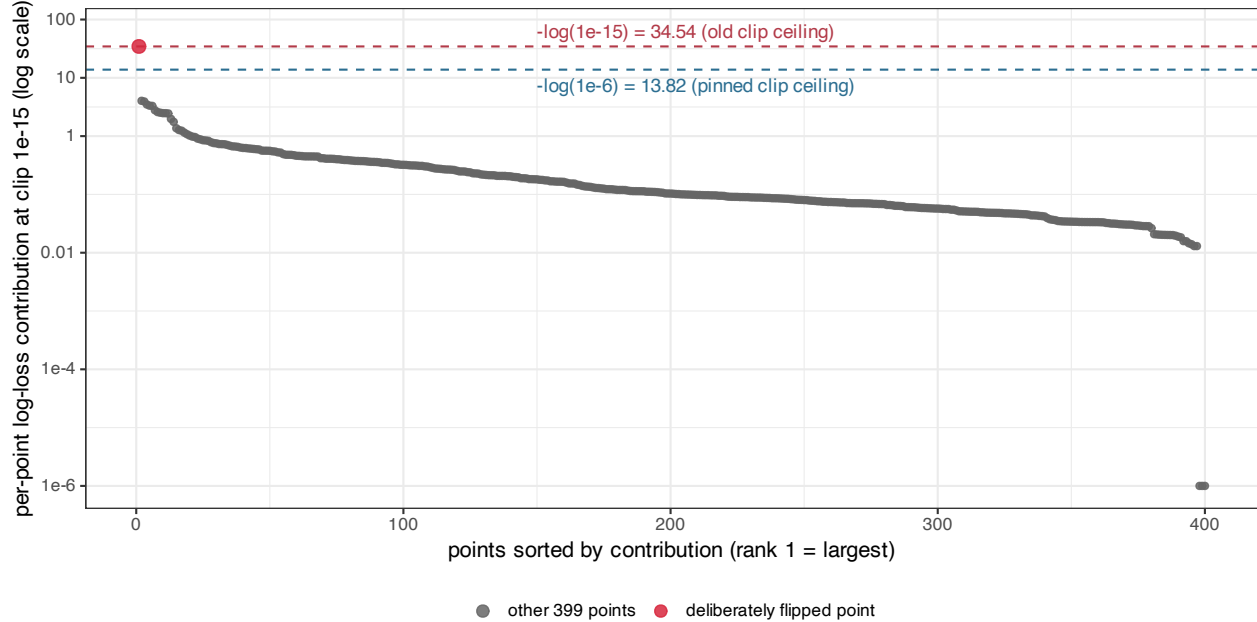
Binomial-mode selection minimizes the cross-validated log loss with probability truncation  $\varepsilon$ :

$$\text{LL}_\varepsilon(j) = -\frac{1}{n} \sum_i \left( y_i \log \text{clip}_{[\varepsilon, 1-\varepsilon]}(\hat{p}_{ij}) + (1 - y_i) \log (1 - \text{clip}_{[\varepsilon, 1-\varepsilon]}(\hat{p}_{ij})) \right).$$

A confidently wrong held-out point ( $\hat{p} \approx 0$ ,  $y = 1$ ) contributes  $-\log \varepsilon = 34.54$  at the previous  $\varepsilon = 10^{-15}$

versus 13.82 at the pinned  $\varepsilon = 10^{-6}$ . On the (b) fixture the deliberately flipped observation’s out-of-fold prediction is exactly 0 under both candidates: its 60-point neighborhoods are all-zero, so the local logistic solve lands in the event-rate fallback, whose weighted mean of zeros is 0.

Fit-status accounting: the binomial cross-validation ran 800 local logistic solves (400 points  $\times$  2 candidates); 785 converged (98.1%) and 15 (1.9%) took the telemetered event-rate fallback — among them all solves for the flipped point. All 800 predictions are finite; none is excluded from any summary.



**Figure 5.** Per-point log-loss contributions at the old truncation  $\varepsilon = 10^{-15}$  for the selected candidate (support 60, degree 0), sorted in decreasing order (log y-axis; the 3 points with contributions below  $10^{-6}$  are drawn at the  $10^{-6}$  display floor). The deliberately flipped point (red, rank 1) contributes  $-\log(10^{-15}) = 34.54$  —  $8.6\times$  the largest other contribution (4.03) — and is the *only* point whose prediction lies outside  $[10^{-6}, 1 - 10^{-6}]$ , so it accounts for 100% of the score change when the truncation moves from  $10^{-15}$  to  $10^{-6}$  (a 0.052 shift of the mean score, both candidates). Pinning  $\varepsilon = 10^{-6}$  caps that single point’s leverage at the blue ceiling; the dashed lines mark the two ceilings.

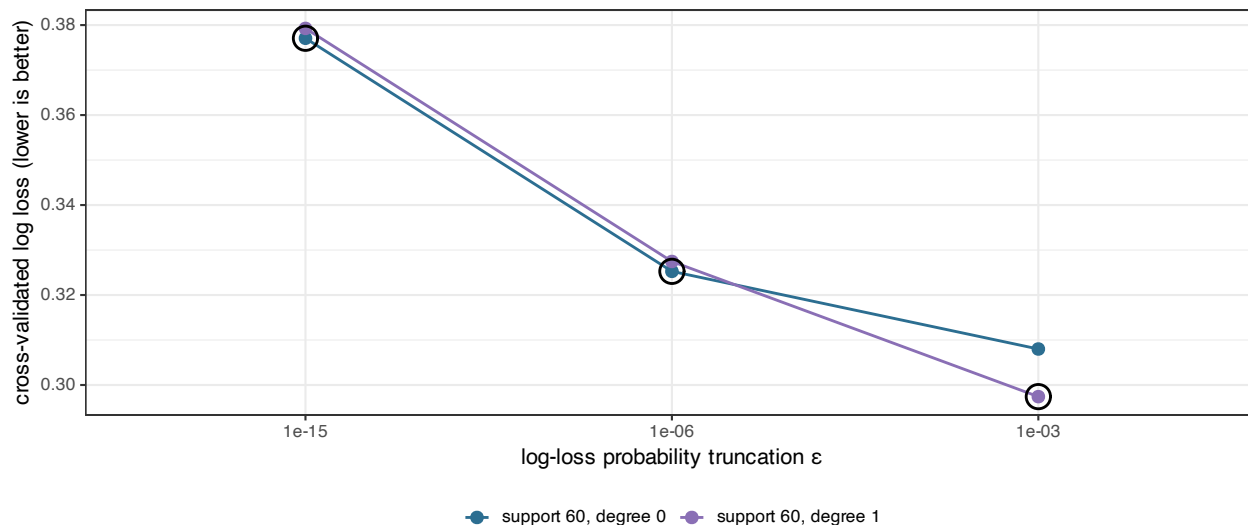
Table 4: Per-candidate log-loss scores at the three truncation levels and the single-point dominance accounting. ‘flip share’ is the fraction of the  $LL(1e-15) - LL(1e-6)$  score change attributable to the flipped observation (1.0 = all of it); ‘flip contrib’ and ‘max other’ compare its per-point contribution at clip 1e-15 with the largest contribution among the other 399 points.  $LL(1e-6)$  equals the deployed selection column `cv.logloss.observed` exactly (residual 0).

| support | degree | LL(1e-15) | LL(1e-6) | LL(1e-3) | flip share | flip contrib | max other |
|---------|--------|-----------|----------|----------|------------|--------------|-----------|
| 60      | 0      | 0.3771    | 0.3253   | 0.3080   | 1.0000     | 34.5388      | 4.0318    |
| 60      | 1      | 0.3793    | 0.3275   | 0.2974   | 1.0000     | 34.5388      | 12.0363   |

## E2.12 study — selection stability across truncation levels

The contract treats cross-clip selection stability as a **study** — its verdict is recorded, never gated, because near-tied candidates may legitimately reselect when the truncation changes. The predeclared rule: the

selection is *stable* iff the log-loss minimizer at  $\varepsilon = 10^{-6}$  equals the minimizer at  $\varepsilon = 10^{-3}$ , computed from the same out-of-fold predictions.



**Figure 6.** Cross-validated log loss of the two E2.12(b) candidates as a function of the truncation level  $\varepsilon$  (black circles mark the minimizer at each level). The selection is identical at  $10^{-15}$  and at the pinned  $10^{-6}$  (support 60, degree 0; margin 0.0022) and *reselects* at  $10^{-3}$  (support 60, degree 1; margin 0.0106): the coarser truncation caps the flipped point’s penalty at  $-\log(10^{-3}) = 6.9$  and also truncates the degree-1 candidate’s confident near-miss predictions, enough to move a 0.002-wide near-tie. The recorded study verdict is therefore “**not stable (near-tie reselected; recorded, not gated)**” (verdict CSV) — precisely the legitimate-reselection case the contract anticipated when it declined to gate this comparison.

## Results summary and discussion

**E2.14 — positive.** On the near-separable fixture the pre-fix solver is demonstrably broken in the way the plan predicted: plain Newton’s deviance increases by  $\approx 489$  at step 2, oscillates, and exhausts the iteration cap, silently degrading the prediction to the event-rate fallback. With step-halving the same support converges in 9 iterations after a single halving, with a monotone trajectory (largest increase  $1.8 \times 10^{-15}$  against the  $10^{-8}$  gate slack), finite coefficients, and  $\hat{p} = 0.125$  strictly inside  $(0, 1)$ . Exact separation behaves exactly as documented: bounded non-convergence at the 50-iteration cap, no NaN, and a deterministic, telemetered fallback under both `unstable.action` modes. The fix engages only where plain Newton misbehaves — on well-behaved solves the accepted iterates are identical, and the Tier-0 E0.6 statistics are bit-identical to the unmodified base commit.

**E2.12(a) — positive.** The train/deploy metric mismatch is not hypothetical: on a sharp G6 surface the raw and deployed Brier scores rank the candidates differently (margins  $> 100\times$  the comparison tolerance), and the pre-fix rule would have deployed the wrong candidate (support 8, degree 0) over the one that is actually better as deployed (support 60, degree 2). Post-fix, the selection column *is* the deployed clipped metric (recomputed-equality residual  $1.4 \times 10^{-17}$ ), so the mismatch class is closed on the R cross-validation path. The legacy C++ cross-validation path still selects on raw aggregate RMSE because it returns no per-point predictions; that scope decision is with the orchestrator (spec-memo item 8).

**E2.12(b) — positive, with the instability precisely localized.** At the old  $10^{-15}$  truncation, one deliberately confident-wrong held-out point contributes 34.54 —  $8.6\times$  any other point — and owns 100% of the score’s sensitivity to the truncation level. The pin at  $10^{-6}$  caps exactly that leverage. The cross-clip study honestly records that a 0.002-wide near-tie reselects at  $10^{-3}$ : cross-clip invariance is a robustness goal, not a correctness criterion, and this fixture sits on the expected boundary.

**Failure modes and residual limits.** Nothing in these results validates the gates against adversarial



mutation — that is the auditor’s mutation-qualification step, deliberately not run by the implementer. All numbers come from one machine/BLAS (macOS arm64, vecLib); the decisive margins ( $\geq 5.8 \times 10^{-4}$ ) sit far above plausible floating-point jitter, but no second platform was exercised. The pre-fix E2.14 trajectory is a committed replication of the pre-change update equations, not an execution of the deleted code. Both fixtures are single deterministic constructions (per the contract’s smoke/full sizing: “deterministic — no replication”), so no uncertainty intervals apply anywhere in this report.

## What we learned

- IRLS instability under near-separation is real but *geometrically selective*: it took an edge-of-cluster flipped label adjacent to a far same-class point to trigger it (a mid-cluster flip is benign). One step-halving is enough to recover a genuine fit that the pre-fix solver abandoned to a fallback — the fix improves fits, not just telemetry.
- The variance floor and  $\pm 35$  linear-predictor clamp already prevented NaN; what they could not prevent was non-monotone oscillation, which is invisible at the endpoint and only assertable on the trajectory. Gating the *trajectory* was the right call.
- Metric mismatches between selection and deployment can flip model choice on realistic sharp-boundary binary surfaces — 16% of out-of-fold predictions leaving  $[0, 1]$  was enough. Selection now optimizes what is deployed.
- A  $10^{-15}$  log-loss truncation converts a single all-zero-neighborhood fallback prediction into the dominant term of the selection score; a  $10^{-6}$  pin bounds that single-point leverage while leaving every in-range contribution untouched.
- A near-tie ( $\Delta \approx 0.002$ ) between candidates can legitimately reselect under truncation changes; recording this as a study verdict rather than gating it keeps the acceptance criteria honest.

## Appendix — reproducibility

All commands assume the package root of worktree `/Users/pgajer/current_projects/geosmooth-t2`, branch `codex/geosmooth-t2-binary-hygiene`.

**Report build:** 2026-06-11 10:20:31 EDT. **Result generation:** 2026-06-11 10:20:19 EDT (UTC 2026-06-11T14:20:19Z) at commit `cb89e12e7c19a38f57f6d5a3e44cc4c0a901b187`; BLAS `libBLAS.dylib`; R version 4.5.2 (2025-10-31). Bundle cross-check at generation: ok vs audit\_artifacts/tier2\_20260611T084642Z: `e2_12a` max `|diff|` = 4.441e-16; `e2_14` traces max `|diff|` = 1.421e-14.

Result artifacts consumed (committed; report-specific inputs under `inputs/` beside this report with SHA-256 in the input manifest; gate/study evidence at the package-level `reports/`):

- E2.14 inputs: `e2_14_fixture.csv`, `e2_14_deviance_traces.csv`, `e2_14_solver_summary.csv`, `e2_14_fallback_telemetry.csv`;
- E2.12 inputs: `e2_12_fixture_a.csv`, `e2_12a_selection_metric.csv`, `e2_12b_contributions.csv`, `e2_12b_clip_dominance.csv`, `e2_12b_fit_accounting.csv`;
- gate/study evidence (`reports/`): `e2_14_prefix_newton_overshoot.csv`, `e2_12_crossclip_scores.csv`, `e2_12_crossclip_stability_verdict.csv`.

Generation and rendering commands:

```
R="dev/methods/lps/reports/tier2/binary_hygiene/2026-06-11"

# 1. (re)generate every report-input CSV + manifest into $R/inputs/
# (deterministic; the optional argument cross-checks against an
# execution bundle):
Rscript "$R/scripts/export_tier2_report_inputs.R" \
  audit_artifacts/tier2_20260611T084642Z
```

```

# 2. pre-fix trajectory + cross-clip study artifacts (committed
#   gate/study evidence under reports/):
Rscript validation/e2_14_prefix_newton_overshoot.R
Rscript validation/e2_12_crossclip_stability_study.R

# 3. render this report, HTML + PDF (written into $R/):
Rscript "$R/scripts/render_lps_tier2_binary_hygiene_report.R"

# acceptance evidence proper (clean committed tree; auditor-facing):
EXECUTOR="<id>" bash scripts/ci/run_tier2_execution_artifact.sh

```

Seeds and determinism: the E2.14 fixtures are seed-free closed-form constructions; E2.12 uses `set.seed(2121)` (sub-item a) and `set.seed(7001)` with the label of observation 7 flipped (sub-item b); cross-validation folds are the explicit `foldid = rep(1:5, length.out = 400)` in all fits (never the internal `cv.seed`).

Related committed documents and machine-local evidence:

- gate test files: `test-lps-binary-separation.R`,  
`test-lps-binary-metric-consistency.R`;
- implementer handoffs: `e2_14_implementer_handoff_2026-06-11.md`,  
`e2_12_implementer_handoff_2026-06-11.md`;
- execution bundles (machine-local, gitignored): `audit_artifacts/tier2_20260611T075822Z` and  
`audit_artifacts/tier2_20260611T084642Z`.